

ENGINEERING CONSTITUTION | VERSION 1.0

EKO EXCEL OS

Build Specification & Engineering Checklist

This document is the single source of truth for the Eko Excel engineering team. Every feature, logic flow, data rule, and acceptance criterion is defined here. If a requirement is not in this document it is not in scope. If it is in this document it must be built exactly as specified. No deviations without written product approval.

Scope: Lagos State public secondary schools | JSS1 – SSS3

Portals: Teacher App · Student Portal · School Admin · District · Ministry

Stack: Flutter (Android) · Next.js · NestJS · PostgreSQL RLS · BullMQ · Claude API

Classification: Confidential | Lagos State Ministry of Education | May 2025



SECTION 00 — READ THIS FIRST

Vision, Objectives & Architecture

• Mission & Core Philosophy

Mission: Build a tripartite education platform connecting Parents, Teachers, and the State (Ministry + Districts) to ensure every child wins in their own game, regardless of background or starting IQ.

Core Loop: Every week runs a Diagnose → Fix → Test cycle. Raw diary data from teachers is processed nightly into WAEC Readiness Meters, Brain Maps, and Hotspot alerts. Dashboards surface interventions. Parents receive WhatsApp reports every Friday at 6PM. Next week's diary data measures improvement.

First Principles: (1) No child is dull. (2) All problems are solved through Diagnose → Fix → Test. (3) Data must be high-quality, low-friction, longitudinal, and role-scoped. (4) Accountability and support must be balanced. (5) Every feature must serve student excellence — nothing else.

- ☐ Every engineer has read and understood the PRD v1.0 in full before writing a single line of code
- ☐ Team agrees: the product is a feedback loop, not a dashboard. Data flows classroom → computation → action → measurement. This order cannot be violated.
- ☐ Team agrees: offline-first is a hard constraint, not a nice-to-have. The teacher app must work with zero connectivity and sync silently when online.
- ☐ Team agrees: row-level security is enforced at the PostgreSQL layer (RLS), not just in the API. Teachers cannot see other teachers' data even if the API is bypassed.

• What Is In & Out of Scope

IN SCOPE

- ☐ Lagos State public secondary schools only — JSS1 through SSS3
- ☐ 5 portals: Teacher App (Android, offline-first), School Admin Dashboard (web + mobile), District Dashboard (web), Ministry Command Center (web), Student Portal (mobile + web)
- ☐ Parent channel: WhatsApp Business API (primary) + SMS fallback
- ☐ AI features: Zero-marking homework engine, 24/7 WhatsApp tutor, predictive stream mapping, Brain Map computation
- ☐ CBT question bank: JSS (12 subjects) + SSS/WAEC (14 subjects), 19,800+ questions minimum
- ☐ QR code attendance system with GPS validation at school gates

- ☐ SmartSpend inventory tracking with digital immutable ledger

OUT OF SCOPE — V1.0

- ☐ Private schools (revenue model post-pilot — not the build target) [OUT OF SCOPE]
- ☐ Primary schools [OUT OF SCOPE]
- ☐ Payment processing or fees collection [OUT OF SCOPE]
- ☐ Video streaming or live class functionality [OUT OF SCOPE]

• Technology Stack — All Decisions Final

Layer	Technology	Justification	Status
Teacher Mobile App	Flutter (Android-first)	Offline-first, SQLite/Hive local storage, low-end Android support	✅ Final
Web Dashboards	Next.js / React.js	School, District, Ministry, Student portals — SSR for performance	✅ Final
Backend API	NestJS (TypeScript)	TypeScript consistency, structured modules, guards for RBAC	✅ Final
Database	PostgreSQL + RLS	Row-Level Security enforced at DB layer. Non-negotiable.	✅ Final
Computation	BullMQ (Redis queues)	Nightly jobs: WAEC Meter, Brain Map, Hotspot detection	✅ Final
Auth	Phone OTP + JWT	No password friction. Teacher logs in with phone number only.	✅ Final
WhatsApp	Meta WhatsApp Business API	Official API only. No unofficial clients.	✅ Final
SMS Fallback	Termii or Africa's Talking	Nigeria-local, reliable delivery — choose one before Phase 1	🟡 Decide
AI Engine	Claude API (claude-sonnet-4-6)	Question generation, WhatsApp tutor, Brain Map insights	✅ Final
File Storage	AWS S3 or Cloudflare R2	Question images, profile photos, reports	🟡 Decide

- ☐ All stack decisions above are final — no substitutions without PM written approval
- ☐ PostgreSQL RLS policies written and tested before any user data is stored [INFRA]
- ☐ Local dev environment documented so any new engineer is running in under 30 minutes [BE]
- ☐ SMS fallback provider selected and API keys stored in secret manager before Phase 4 [INFRA]

• Role Hierarchy & Data Scoping Rules

Non-Negotiable Security Rule: Every data query in every portal must be filtered by the requesting user's role and their associated school_id / district_id / ministry scope. There is no admin bypass mode in production. No exceptions.

Role	Data Scope	Student PII Access	Can Message
Teacher	Own classes only	Yes — own students only	Receives nudges; logs notes
School Admin	Entire school	Yes — all school students	Parents + teachers
District Officer	All schools in district	Aggregated only (no names)	Broadcast to schools/parents
Ministry / SUBEB	State-wide aggregates	No individual PII	Statewide broadcast
Parent	Own child only (WhatsApp)	Own child only	Reply to Eko Excel bot
Student	Own profile only	Own data only	Interact with AI tutor

- ☐ PostgreSQL RLS policy written for each table × role combination
- [INFRA]
- ☐ NestJS guards enforce role check on every protected endpoint — no endpoint is unguarded
- [BE]
- ☐ Test: teacher cannot retrieve another teacher's diary even with a valid JWT and direct API call
- [TEST]
- ☐ Test: district officer cannot retrieve individual student names — only counts, percentages, and anonymised IDs
- [TEST]

• Build Phase Priority Order

STEP 1

Phase 1 — MVP Core (Weeks 1–8)

Teacher App (Flutter Android) · Daily diary entry with attendance, topic, score, 5 behavioural traits · Backend data model · PostgreSQL RLS · OTP auth · WAEC Readiness Meter calculation · Basic school admin dashboard
→ Acceptance: Teacher submits a diary in under 2 minutes on a Tecno Spark

STEP 2

Phase 2 — Intelligence (Weeks 9–16)

Student Profile · Brain Map computation engine · Nightly BullMQ jobs · Hotspot detection · Full school dashboard · Student portal (read-only profile)
→ Acceptance: Brain Map refreshes automatically each Sunday at 2AM WAT

STEP 3

Phase 3 — Scale (Weeks 17–24)

District dashboard · Ministry Command Center · Advanced Google-like search · LGA heatmap · Accountability + nudge workflows · AI homework engine · CBT question bank
→ Acceptance: Ministry officer sees statewide readiness in under 10 seconds

STEP 4

Phase 4 — Engagement (Weeks 25–32)

WhatsApp Business API · Automated Friday parent reports · 24/7 AI WhatsApp tutor · SMS fallback · QR code gate check-in · SmartSpend inventory · Full polish + security audit
→ Acceptance: Parent receives automated Friday 6PM WhatsApp report with correct child data

SECTION 01 — TEACHER PORTAL

Teacher Mobile App — Android-First, Offline-First | Phase 1 Priority

Objective: Enable every Lagos public school teacher to submit a complete daily classroom diary — attendance, topic covered, class performance score, and 5 behavioural trait ratings — in under 2 minutes, on any low-end Android device, with or without internet connectivity.

Hard Constraint: The 2-minute target is non-negotiable. Every UI decision must be tested against this constraint. If a teacher has to type anything (other than an optional note), the design has failed.

• Authentication & Onboarding

- ☐ Phone number entry accepts 0XX and +234XX formats interchangeably — normalised to +234 internally [FE]
- ☐ OTP sent via SMS: 6-digit, expires in 5 minutes, max 3 retries before 15-minute lockout [BE]
- ☐ On first login: teacher sees their pre-loaded class list (seeded from admin — zero manual entry required) [FE]
- ☐ JWT stored securely in device keystore (not SharedPreferences). Refresh token valid for 30 days. [BE]
- ☐ Offline mode: if no connectivity on launch, app opens from local SQLite cache and shows last-synced state [FE]
- ☐ Biometric unlock (fingerprint) after first OTP login — skip OTP on subsequent sessions if biometric passes [FE]
- ☐ Single onboarding screen (2 taps max) if profile incomplete — never blocks diary access [FE]

• Home Dashboard Screen

- ☐ Greeting with teacher first name + time-of-day salutation (Good morning / afternoon / evening) [FE]
- ☐ Today's class list shown immediately — sorted by timetable period. Each card shows: class name, subject, period, diary status (Pending / Submitted / Missed) [FE]
- ☐ Status colour coding: Pending = Lagos Yellow (#FFCC00), Submitted = Lagos Green (#008751), Missed = Lagos Red (#E30613) [FE]
- ☐ Notification bell shows count of unread messages from school admin or district [FE]

- ☐ Weekly personal impact strip: 'You submitted X of Y diaries this week. Class readiness: Z%' [FE]
- ☐ Sync status indicator showing last synced timestamp — shows 'Offline mode' banner if no connectivity [FE]
- ☐ Hotspot alert badge: 'X students need attention' — taps through to hotspot list [FE]
- ☐ Bottom navigation: Home / My Classes / Diary / Students / Messages — 5 tabs, fixed, always visible [FE]
- ☐ Tapping a class card opens Diary Entry screen pre-populated with that class + today's date [FE]
- ☐ All home screen data loads from local SQLite first, then refreshes from API in background — no loading spinner on open [FE]

• Diary Entry Screen — MOST CRITICAL SCREEN

Design Rule: Every input must be click/tap-only. Zero text typing required for core fields. Pre-loaded dropdowns only. If a teacher must type to complete a diary, the build has failed.

STEP 1 — ATTENDANCE (TARGET: 30 SECONDS)

- ☐ Student list pre-loaded from database — sorted alphabetically. Shows full name + initials avatar. [FE]
- ☐ Default state: all students PRESENT (green). Teacher taps to mark ABSENT (red). Faster than marking each individually. [FE]
- ☐ Attendance grid: 5 students per row, circles with initials — optimised for speed [FE]
- ☐ Running count displayed: '26 Present / 6 Absent' — updates on each tap in real time [FE]
- ☐ QR code attendance mode (Phase 4): students scan class QR on teacher's phone to self-register presence [FE (Ph4)]
- ☐ Attendance stored locally in SQLite immediately — not lost if app crashes before full submission [FE]

STEP 2 — TOPIC COVERED (TARGET: 15 SECONDS)

- ☐ Topic picker: pre-loaded Lagos NERDC curriculum topics for this class × subject. Multi-select (up to 3 topics per session). [FE]
- ☐ Topics displayed as scrollable chips — not a dropdown. Teacher taps chip to select, taps again to deselect. [FE]
- ☐ WAEC flag shown next to topics with waec_weight ≥ 7 — visual cue only, not mandatory [FE]

- ☐ Optional free-text note field (max 200 chars) — explicitly labelled 'Optional note' [FE]

STEP 3 — CLASS PERFORMANCE SCORE (TARGET: 10 SECONDS)

- ☐ Horizontal slider: 0–100. Default position: 60. Large drag handle for fat-finger use. Current value in large text above. [FE]
- ☐ Colour transitions: 0–54 = red zone, 55–74 = yellow zone, 75–100 = green zone. Matches WAEC thresholds. [FE]
- ☐ Score is class-level aggregate — not per student. Feeds WAEC Readiness Meter as Normalized_Recent_Score_i. [LOGIC]

STEP 4 — 5 BEHAVIOURAL TRAITS (TARGET: 25 SECONDS)

- ☐ All 5 traits on one screen — not paginated: Engagement/Participation, Persistence/Effort, Focus/Attention, Collaboration, Resilience/Emotional Regulation [FE]
- ☐ Rating input: 5 tappable circles (1–5) per trait. Pre-filled at 3 (neutral). Teacher adjusts outliers only — reduces total taps. [FE]
- ☐ Trait ratings are class-level (how the whole class performed on this trait) — not per-student at diary entry stage [LOGIC]
- ☐ Long-press any trait name shows 1-sentence rubric definition for each score level (from PM rubric document) [FE]

STEP 5 — SUBMIT

- ☐ Single large 'Submit Diary ✓' button — Lagos Green. Full-width. Cannot be missed. [FE]
- ☐ On submit: immediate success animation + 'Class readiness updated tonight' message. Returns to home after 1.5 seconds. [FE]
- ☐ If offline: diary saved to SQLite with status = 'pending_sync'. Syncs automatically on reconnect. Teacher never sees an error. [FE]
- ☐ Submission timestamp stored in UTC, converted to WAT (UTC+1) for display — critical for nudge timing calculations [BE]
- ☐ Raw diary data is immutable after submission — no editing allowed after 11:59PM on the submission day [LOGIC]

• Student Profile View (Teacher-Facing)

- ☐ Student header: photo/avatar, full name, class, student ID, age. WAEC Readiness % badge + Attendance % badge always visible. [FE]

- ☐ 4 tabs: Overview · Brain Map · Academic Trends · History [FE]
- ☐ Overview tab: WAEC Readiness donut chart per subject, attendance summary, current hotspots list, last diary entry details [FE]
- ☐ Brain Map tab: 5-domain horizontal bar chart. Top 2–3 strength profiles as chips. Teaching recommendation text from AI (refreshed weekly). [FE]
- ☐ Brain Map shown only from Week 5 onward — insufficient data message before then with progress indicator [LOGIC]
- ☐ Academic Trends tab: line chart showing WAEC Readiness % over time per subject. Minimum 6 weeks of data. [FE]
- ☐ History tab: chronological diary timeline — date, topic, score, trait scores, attendance status for each session [FE]
- ☐ Action strip: 'Message Parent' (triggers WhatsApp template) · 'Log Intervention' (text note stored against student record) [FE]
- ☐ Teacher can only access profiles of students in their own classes — RLS enforced at database layer [LOGIC]
- ☐ Hotspot badge (red) shown on student card in class list if any subject readiness < 55% [FE]
- ☐ Stream Prediction card (Phase 3+): AI-recommended stream (Sciences / Arts / Commercial) with confidence % and evidence summary. Visible from JSS3 Term 2 onward. [FE (Ph3)]
- ☐ All student profile data loaded from API — personal data does not persist on device beyond session [LOGIC]

• Hotspots, Messages & Nudge Handling

- ☐ Hotspots screen: student + topic combinations where readiness < 55% (Red) or 55–74% (Yellow). Sortable by severity. Each row shows student name, subject, current %, trend direction (↑ ↓ →). [FE]
- ☐ Messages screen: inbox for queries from school admin or district. Sender role, subject, timestamp, and read receipt required. [FE]
- ☐ Teacher can reply to messages — plain text, 500 char limit. Replies logged with timestamp. [FE]
- ☐ Nudge push notification: sent at 4PM if no diary submitted for any class that day. Deep links directly to diary entry. [BE]

- ☐ Second nudge: WhatsApp message at 6PM if still not submitted after first push notification [BE]
- ☐ If not submitted by 11:59PM: status marked 'Missed' and escalates to school admin compliance dashboard next morning [LOGIC]
- ☐ Teacher can submit a reason for missed entry (dropdown: School closed / Sick leave / Power outage / Network issue / Other). Pauses escalation but does not remove missed record. [FE]
- ☐ Support request button: teacher can flag a query + support request to district. Routed to district accountability flow. [FE]

• AI Homework Engine — Teacher-Side (Phase 3)

Objective: Teacher enters 3 inputs (Topic, WAEC Difficulty level, Question Count). AI generates a standardised assessment, delivers to students via student portal or printed QR, marks all responses, and returns a diagnostic report. Teacher does zero marking.

- ☐ Homework creation: Topic picker (same as diary), Difficulty selector (1=Easy / 2=Medium / 3=Hard/WAEC), Question Count (5/10/15/20) [FE]
- ☐ AI pulls questions from Postgres question bank first. If bank has < 5 matching questions, calls Claude API to generate and stores new questions back in the bank. [AI]
- ☐ Generated assessment assigned to entire class with teacher-set deadline (tonight / tomorrow / this week) [FE]
- ☐ Students complete via Student Portal CBT interface or WhatsApp bot (multiple choice, max 5 questions via WhatsApp) [FE]
- ☐ System marks all responses automatically — no teacher action required after creation [BE]
- ☐ Teacher receives diagnostic summary: class average, lowest 3 students, most-missed question (topic gap signal) [FE]
- ☐ Homework results feed directly into WAEC Readiness Meter — each completed assessment updates student's topic score [LOGIC]
- ☐ Teacher can print assessment as a QR-coded paper sheet — students scan to submit answers on their phones [FE]
- ☐ Estimated time saving displayed: 'This would take you ~X hours to mark manually. We did it in 0 seconds.' [FE]



SECTION 02 — STUDENT PORTAL

Student Portal — Mobile + Web | Phase 2–3

• Student Dashboard — Own Profile

- ☐ Student logs in via phone OTP (own number or parent's number registered to their record) [BE]
- ☐ Home screen: personalised greeting, current WAEC Readiness % donut, attendance %, and a 'Today's Action' card (1 specific study action) [FE]
- ☐ Subject Readiness grid: all 5 core subjects shown as coloured progress rings. Tap each to drill into topic breakdown. [FE]
- ☐ Brain Map card: Top 3 strength profiles in plain language ('You learn best through logic and patterns — work through problems step by step') [FE]
- ☐ Active Assignments section: homework set by teacher — shows subject, topic, deadline, and completion status [FE]
- ☐ CBT Practice button: shortcut to question bank pre-filtered for student's class level and weakest subject [FE]
- ☐ AI Tutor button: deep-links to WhatsApp chat with Eko Excel AI tutor bot [FE]
- ☐ Attendance history: calendar view — green = present, red = absent, grey = holiday/weekend [FE]
- ☐ Stream Prediction banner (JSS3 Term 2+): AI-generated recommendation with confidence % and top 3 evidence points [FE (Ph3)]
- ☐ Student can NEVER see other students' data — own profile only. RLS enforced at database layer. [LOGIC]
- ☐ Class leaderboard: shown only if school admin enables it — relative percentile, not absolute position [FE]

• CBT Practice Engine — WAEC & JSCE Prep

Objective: Student can practise WAEC-style CBT questions by subject and topic at any time. System tracks all responses, updates WAEC Readiness Meter, and surfaces the exact topics needing attention. No passive reading — active retrieval practice only.

- ☐ Practice mode: student selects Subject → Topic → Difficulty → Question Count → Start. Pre-filters show estimated time. [FE]

- ☐ Timed mode: WAEC simulation — 50 questions, 90 minutes, full-screen, no back navigation. Timer prominent. [FE]
- ☐ Question display: stem text, 4 options (A/B/C/D) as full-width tap targets. Submit button appears after selection — no accidental submissions. [FE]
- ☐ After each question (practice mode): instant feedback — correct = green, wrong = red with correct answer highlighted. Full explanation displayed. [FE]
- ☐ Timed/WAEC simulation mode: no instant feedback during exam. Full feedback shown after paper submission. [FE]
- ☐ Image support: questions with diagrams (Biology, Physics, Technical Drawing) display embedded image from S3. Image loads before question is answerable. [FE]
- ☐ Session results screen: score, percentage, time taken, correct/incorrect by topic, weakest topic highlighted, 'Practice this topic' shortcut [FE]
- ☐ Every completed session recorded in student_sessions: question_id, response, is_correct, time_spent_seconds, source, timestamp [BE]
- ☐ Session results feed into WAEC Readiness Meter calculation nightly — same formula as classroom diary scores [LOGIC]
- ☐ Adaptive question selection: if student scores < 55% on a topic, next session for that topic starts with easier questions (difficulty=1) [LOGIC]
- ☐ WAEC past paper filter: toggle 'Past WAEC Questions Only' — filters to questions with waec_year populated [FE]
- ☐ Progress streak: 'You've practised 5 days in a row' — motivational only, not tied to grading [FE]
- ☐ Offline CBT: question bank syncs locally. Student can practise with zero connectivity. Results sync when online. [FE]
- ☐ Question never repeated in same session. Questions rotate using seen_questions array stored per student. [LOGIC]

• 24/7 AI WhatsApp Tutor — Student-Facing

Objective: A student sends any academic question to the Eko Excel WhatsApp number. The AI responds with a contextual, Lagos-relevant explanation, works through the solution step by step, and optionally quizzes the student to confirm understanding. Available 24/7.

- ☐ Student sends question as WhatsApp text message to registered Eko Excel Business number [API]

- ☐ System identifies student by WhatsApp number. If unregistered: bot replies 'Please ask your teacher to register your number.' [BE]
- ☐ Claude API called with system prompt including: student's class level, weakest subjects from Brain Map, Lagos curriculum context. Age-appropriate response. [AI]
- ☐ Explanations use locally relevant analogies (Lagos traffic, market pricing, lagoon depth for physics problems) [AI]
- ☐ After explanation: bot asks 'Do you want to try a practice question on this topic? Reply YES or NO' [BE]
- ☐ If YES: sends 1 multiple-choice question. Student replies with letter (A/B/C/D). Bot confirms correct/incorrect with brief explanation. [BE]
- ☐ All tutor interactions logged in student_tutor_sessions: question_text, response_text, topic_detected, timestamp. Feeds Brain Map. [BE]
- ☐ Rate limit: max 20 tutor exchanges per student per day to prevent abuse and control Claude API costs [BE]
- ☐ Guardrail: if non-academic message sent, bot replies: 'I'm here to help with schoolwork! Ask me anything about your subjects.' No Claude API call made. [LOGIC]



SECTION 03 — SCHOOL ADMIN PORTAL

Headteacher & School Admin Dashboard — Web + Mobile | Phase 1–2

• School Overview Dashboard

- ☐ 4 headline KPI tiles (always visible): School WAEC Readiness %, Teacher Diary Compliance %, Active Hotspot Count, Total Students Tracked [FE]
- ☐ Readiness trend sparkline on each KPI tile — 8-week mini chart showing direction of travel [FE]
- ☐ Teacher compliance table: each teacher row shows Name, Subjects, Classes, This Week's Submissions, Rate %, Status badge (Compliant / Warning / At Risk) [FE]
- ☐ One-click 'Send Nudge' per teacher row — sends WhatsApp message. Admin can customise or use default template. [FE]
- ☐ One-click 'Send Query' per teacher — structured query form (issue, severity, required response date). Logged in accountability trail. [FE]
- ☐ Subject performance heatmap: grid of Subject × Class — each cell colour = average WAEC Readiness. Red/Yellow/Green. Click cell → student list. [FE]
- ☐ Top Hotspots panel: top 10 student-subject combinations with readiness < 55%, sorted by severity. Student name, subject, current %, trend. [FE]
- ☐ Brain Map Distribution chart: school-level aggregate of cognitive domain scores — policy insight for teaching strategy [FE]
- ☐ Global search: type any student name, class, or subject → instant filtered results across all school data [FE]
- ☐ Message Parents button: broadcast WhatsApp + SMS fallback message to all parents in a class or school-wide [FE]
- ☐ Intervention assignment: from any hotspot, admin assigns support action to a teacher with a due date. Tracked in intervention_log. [FE]
- ☐ QR morning gate dashboard (Phase 4): live teacher gate check-in status — who has arrived, who is absent, time of check-in [FE (Ph4)]

• School Admin Operations — Data Management

- ☐ Student roster management: add / edit / archive students. Bulk CSV upload. Fields: name, DOB, class, gender, parent_phone, student_id. [FE]

- ☐ Class management: create/edit class groups, assign teachers to class+subject combos, set academic term dates [FE]
- ☐ Teacher accounts: invite by phone number, set class+subject assignments, deactivate/reactivate. Zero manual entry on teacher side. [FE]
- ☐ Timetable import: upload weekly timetable as CSV or configure in-app grid. Drives 'today's classes' view in teacher app. [FE]
- ☐ School profile: name, LGEA, district, address, headteacher name + contact, school_id (immutable, set by Ministry) [FE]
- ☐ Term calendar: set first/last day per term, public holidays, school closure days. Excluded from diary compliance calculations. [FE]
- ☐ Report export: download school-level WAEC Readiness report as PDF for any date range. Auto-formatted with school header and MoE logo. [FE]
- ☐ SmartSpend inventory (Phase 4): add/edit/track physical assets. Each item has a digital record with condition, location, responsible person. [FE (Ph4)]
- ☐ Inventory alert: admin notified when item reported missing or condition changes to 'damaged/lost'. Audit trail stored immutably. [BE (Ph4)]
- ☐ NDPR consent tracking: flag students missing parent consent. Their data visible to school but not to district/ministry levels. [LOGIC]



SECTION 04 — DISTRICT PORTAL

District Officer Dashboard — Web-First, Advanced Search | Phase 3

• District Overview Dashboard

- ☐ 5 headline KPIs: District Avg WAEC Readiness %, Teacher Compliance %, Total Hotspots, Schools Below 55% Threshold, Active Interventions Count [FE]
- ☐ School comparison table: rows = schools. Columns = Readiness %, Compliance %, Hotspot Count, Trend (↑↓→), Last Diary Submitted. Sortable by each column. [FE]
- ☐ Click any school row → drill into School Detail view (same data as school admin, read-only for district officer) [FE]
- ☐ District Hotspot Map: schools on Lagos map. Markers coloured by readiness level. Cluster view for dense areas. [FE]
- ☐ Subject Performance Heatmap: rows = schools, columns = subjects. Cell = average readiness for that subject in that school. [FE]
- ☐ Week-over-week trend chart: district average readiness over 12 weeks, broken down by subject. One line per subject. [FE]
- ☐ Teacher compliance leaderboard: all teachers in district ranked by submission rate. Top 10 and bottom 10 visible. [FE]
- ☐ District officer can send structured query to any school — issue, evidence, required action, deadline. Logged in accountability trail. [FE]
- ☐ Intervention dispatch: deploy support action (teacher training, resource allocation) to one or multiple schools. Track completion. [FE]
- ☐ Broadcast message to parents: district-wide WhatsApp to all parents (filter by school/class/subject). Template pre-approval required. [FE]
- ☐ Monthly report auto-generation: district PDF compiled on 1st of each month — sent to district officer + Ministry automatically [BE]

• Advanced Search — Google-Like District Query

Objective: A district officer must be able to type 'Low Persistence students in Ikorodu' or 'SS2 Math weak in Trigonometry' and get an immediate, accurate filtered result across all schools in their district.

- ☐ Search bar always visible at top of district dashboard. Full-text search across student Brain Map profiles, subject readiness, school names, teacher names. [FE]
- ☐ Structured filter panel: School (multi-select), Class Level (JSS1–SSS3), Subject, Readiness Range (slider 0–100%), Behavioural Trait + Score Range [FE]
- ☐ Results returned in under 2 seconds for any filter combination across up to 50 schools — performance SLA enforced in CI/CD [BE]
- ☐ Results show: Student Name, School, Class, Matching Metric Value, Trend. Row click opens student profile (read-only, district-scoped). [FE]
- ☐ Save search as 'Watchlist' — district officer gets weekly digest of how this cohort is trending. Max 5 saved watchlists per officer. [FE]
- ☐ Export results as CSV (student_id, school_id, class, metrics — no names) or PDF (names included). Privacy-appropriate formats. [FE]
- ☐ PostgreSQL search indexes: tsvector on topic names and Brain Map labels, GiST index on readiness scores, B-tree on school_id and class [BE]
- ☐ Search returns ONLY students within the district officer's district_id scope — cannot return students from other districts [LOGIC]

SECTION 05 — MINISTRY GOVERNANCE CENTER

Ministry / SUBEB / MoE — State-Level Command Dashboard | Phase 3

Design Requirement: A Permanent Secretary or Director who opens the Ministry dashboard must understand the state of Lagos public secondary education in 10 seconds without clicking anything. If this understanding requires more than 10 seconds or one click, the design has failed. This is a hard acceptance criterion.

Ministry Data Rule: No individual student names or teacher names visible at Ministry level. All data is aggregated at school or district level minimum. Exceptions only for anonymised research exports approved by the PS office.

• Statewide Command Center Features

- ☐ State Readiness Gauge: single large number — statewide average WAEC Readiness % with week-over-week delta (e.g. '71% ↑+3%'). Most prominent element on page. [FE]
- ☐ 5 headline KPI tiles: State Readiness %, Teacher Submission Rate (target ≥90%), Active Hotspots Statewide, Total Students Tracked, Parent WhatsApp Open Rate [FE]
- ☐ Lagos LGA Heatmap: colour-coded map of all LGAs by average readiness — Red (<55%), Yellow (55–74%), Green (≥75%). Hover shows LGA name, readiness %, school count. [FE]
- ☐ District comparison bar chart: all 5 Lagos education districts ranked by readiness %. Bars coloured by threshold. Click bar → drill into that district (read-only). [FE]
- ☐ Year-over-year trend chart: state average readiness by term for last 3 academic years. Ministry's primary strategic chart. [FE]
- ☐ Subject Weakness Ranking: subjects ranked 1–14 by lowest average readiness statewide. Drives curriculum intervention decisions. [FE]
- ☐ Brain Map Distribution: statewide cognitive profile aggregate. % of students Logical-dominant vs Verbal-dominant etc. Policy insight for teacher training allocation. [FE]
- ☐ Teacher Compliance Heatmap: district × week grid. Cell colour = compliance rate. Instantly shows sustained compliance problems. [FE]
- ☐ Broadcast Message: compose and send targeted WhatsApp + SMS to all Lagos parents (filter by district/subject/readiness threshold). Pre-approved template required. [FE]
- ☐ Hotspot Resolution Rate metric: % of hotspots flagged 4 weeks ago that have improved above 55%. Target: ≥50% resolution per term. Displayed prominently. [FE]
- ☐ Gender equity panel: readiness broken down by gender across subjects. Flags subjects with ≥10% gender gap. Drives targeted intervention. [FE]

- ☐ QR Gate Compliance (Phase 4): % of teachers GPS-verified present at school gates each morning, statewide. Breakdown by district and school. [FE (Ph4)]
- ☐ SmartSpend Overview (Phase 4): total items registered statewide, items reported missing this term, schools with highest loss rates [FE (Ph4)]
- ☐ Automated weekly briefing: every Monday 7AM, Ministry users receive WhatsApp summary — state readiness delta, top 3 district concerns, compliance rate [BE]
- ☐ Anonymised aggregate CSV export for international partners (UNESCO/UNICEF) — accessible to PS-level role only [FE]
- ☐ Audit log: every Ministry-level action logged with user, timestamp, and action. Immutable. Accessible to PS role only. [BE]



SECTION 06 — PARENT CHANNEL

WhatsApp Business API + SMS Fallback | Phase 4**• Automated Friday Report Flow****STEP 1****Friday 5:58PM — Report Generation Job Fires**

BullMQ job runs. For each student with registered parent_whatsapp_number: pulls this week's attendance, subject readiness scores, top focus area, 1 home action, Brain Map insight (if ≥5 weeks of data). Constructs personalised message payload.

→ **Trigger: cron job every Friday at 17:58 WAT**

STEP 2**6:00PM — WhatsApp Template Message Sent**

Meta WhatsApp API called with pre-approved template ID. Variables injected: parent_name, student_name, attendance_pct, 5 subject scores with emoji, focus_area, home_action. If WhatsApp delivery fails within 60s → SMS fallback triggered.

→ **Requires: pre-approved Meta message template**

STEP 3**Parent Receives Message & Replies**

Reply options: MORE · ATTENDANCE · BRAIN MAP. Free text also accepted. Any parent reply triggers webhook handler.

→ **Any reply → webhook handler in under 5 seconds**

STEP 4**Reply Router Logic**

MORE → detailed subject breakdown. ATTENDANCE → full term calendar. BRAIN MAP → top 3 profiles + home recommendations. Free text → Claude API classifies as academic question (AI tutor) or school concern (routed to school admin inbox).

STEP 5**School Admin Receives Routed Messages**

School concern messages appear in school admin Messages inbox within 60 seconds. Admin or teacher can reply — delivered back to parent via WhatsApp.

- ☐ WhatsApp Business account registered, verified, and Meta-approved BEFORE pilot day 1. [INFRA]
Display name: 'Eko Excel Official'. Category: Education.
- ☐ All 8 message templates submitted to Meta for pre-approval minimum 5 business days before first send. Templates cannot be edited after approval. [INFRA]
- ☐ SMS fallback: if WhatsApp delivery status = 'failed' after 60s → SMS API sends truncated version (max 320 chars) to same number [BE]
- ☐ Delivery status tracked: each message logged with status = sent / delivered / read / failed. Parent engagement rate reported on Ministry dashboard. [BE]
- ☐ Opt-out handling: parent replies STOP → number added to opt_out table. No further messages sent. Compliant with NDPR and Meta policy. [LOGIC]

- ☐ Parent with multiple children: each child gets a separate message. Parent can use student initials as prefix for replies (e.g. 'AK BRAIN MAP'). [LOGIC]
- ☐ Rate limiting: max 1 automated outbound message per parent_number per day to avoid WhatsApp spam classification [BE]



SECTION 07 — DATABASE & COMPUTATION

Core Data Model, WAEC Formula & Brain Map Logic

• Core Database Tables

Table	Key Fields	Notes
schools	school_id, name, district_id, lga, address, headteacher_id	Seeded from MoE master list. school_id is immutable.
districts	district_id, name, lga_list (array), officer_id	5 Lagos education districts.
users	user_id, phone, role (enum), school_id, district_id, is_active	RLS policies key off role + school_id / district_id
students	student_id, school_id, class_id, name, dob, gender, parent_whatsapp, consent_signed	Immutable student_id. parent_whatsapp nullable.
classes	class_id, school_id, level (JSS1–SSS3), section (A/B/C), academic_year	New record each academic year.
teacher_class_subjects	teacher_id, class_id, subject_id	Junction table. Teacher sees only their classes.
diary_entries	diary_id, teacher_id, class_id, subject_id, date, attendance_json, topics_covered[], class_score, traits_json, submitted_at	IMMUTABLE after 11:59PM submission day. Raw source of truth.
student_attendance	record_id, student_id, diary_id, date, status (present/absent/excused)	Derived from attendance_json on diary sync.
topic_segments	topic_id, subject_id, class_level, topic_name, sub_topic, waec_weight (1–10)	Pre-loaded. Never changed by users. PM-owned content.
computed_waec_meters	meter_id, student_id, subject_id, readiness_pct, computed_at, contributing_entries[]	Refreshed nightly. Written by BullMQ job ONLY.
brain_maps	map_id, student_id, logical_pct, spatial_pct, verbal_pct, applied_pct, consistency_pct, top_profiles[], recommendations, computed_at	Refreshed weekly Sundays 2AM. Written by BullMQ job ONLY.
hotspots	hotspot_id, student_id, subject_id, readiness_pct, detected_at, resolved_at, status	Auto-created when readiness < 55%. Resolved when ≥65% for 2 consecutive weeks.
questions	question_id, subject_id, class_level, topic_id, stem, options_json, correct_answer, explanation, difficulty, waec_year, waec_topic_weight, image_url	19,800+ rows. PM-owned content.
student_sessions	session_id, student_id, question_id, response, is_correct, time_spent_secs, source (cbt/whatsapp/homework), timestamp	Every answer attempt logged. Feeds WAEC Meter.
whatsapp_messages	msg_id, phone, direction (in/out), content, template_id, status, delivered_at, read_at	Full delivery audit trail. Purged after 18 months per NDPR.
interventions	intervention_id, target_type, target_id, assigned_by, assigned_to, description, due_date, status, completed_at	Accountability trail. Immutable after completion.
inventory_items	item_id, school_id, category, description, condition, responsible_user_id, status	Phase 4. SmartSpend ledger.

• WAEC Readiness Formula — Implementation Specification

Formula: $\text{Readiness_}\% = [\text{Sum}(\text{Topic_Weight_i} \times \text{Normalized_Recent_Score_i} \times \text{Recency_Factor_i})] \times \text{Adjustment_Factors}$
 $\text{Topic_Weight_i} = \text{waec_weight from topic_segments (1–10, PM-defined from past-paper frequency analysis)}$
 $\text{Normalized_Recent_Score_i} = (\text{score} / 100)$ weighted average of diary class_score + CBT session results for that topic
 $\text{Recency_Factor_i} = e^{(-0.1t)}$ where $t = \text{days since data point}$. $\text{Lambda} = 0.1/\text{month}$ as starting value (tunable per subject post-pilot)
 $\text{Adjustment_Factors} = \text{attendance multiplier (readiness capped at 60\% if attendance} < 60\%) \times \text{data_confidence (weight halved if fewer than 3 data points for a topic)}$
 Output: 0–100% integer. Thresholds: Green $\geq 75\%$, Yellow 55–74%, Red $< 55\%$.

- ☐ Computation runs as BullMQ job every night at 2AM WAT for all students with new diary data in last 7 days [BE]
- ☐ All computation inputs read from immutable diary_entries and student_sessions — never modified post-submission [LOGIC]
- ☐ Results written to computed_waec_meters with timestamp. Previous results retained for trend charts — never overwritten. [BE]
- ☐ Lambda (λ) stored as a config variable per subject in database — updatable by PM without code deployment [BE]
- ☐ Hotspot detection: after each nightly run, any student-subject where readiness_pct $< 55\%$ and no open hotspot → new hotspot record created. Auto-notifies school admin. [BE]
- ☐ Hotspot resolution: if readiness $\geq 65\%$ on 2 consecutive weekly computations → hotspot.status = 'resolved'. Auto-notifies school admin. [BE]

• Brain Map Computation — Implementation Specification

Brain Map = 75% Academic + 25% Behavioural (exact split — final): 5 Cognitive Domains: Logical-Analytical: Mathematics + Physics academic scores Spatial-Mechanical: Physics (optics/mechanics) + Technical Drawing scores Verbal-Creative: English Language + Literature scores Applied-Practical: Agricultural Science + Home Economics + Chemistry (practical) scores Consistency: Attendance rate x diary submission consistency (Persistence trait average) Behavioural mapping: Engagement → all domains. Persistence → Consistency. Focus → Logical-Analytical. Collaboration → Verbal-Creative. Resilience → Applied-Practical. Output: Score 0–100% per domain. Top 2–3 domains = strength profiles. Claude API generates teaching recommendations using top profiles as context.

- ☐ Brain Map runs weekly — every Sunday at 2AM WAT. Requires minimum 3 weeks of diary data to generate first output. [BE]
- ☐ Minimum data gate: fewer than 3 weeks of data → Brain Map shows 'Insufficient data — available in X weeks'. No partial output displayed. [LOGIC]
- ☐ After computation: Claude API called with domain scores to generate plain-English profiles, teaching recommendations, and home actions. Stored as JSON in brain_maps.recommendations. [AI]

- ☐ Stream Prediction (Phase 3, JSS3 Term 2+): Claude API called with full Brain Map history to recommend Sciences / Arts / Commercial with confidence % and top 3 evidence points. [AI]
- ☐ All Brain Map computations stored longitudinally — every weekly result retained. Trends shown from JSS1 through SSS3. [BE]



SECTION 08 — THE CONSTITUTION

12 Non-Negotiable Engineering Laws — Read Before Writing Any Code

Status: These 12 laws cannot be overridden by any sprint, release pressure, or technical convenience. Violations are rejected at code review. Engineers who disagree raise it to the PM in writing before building — not after.

L Offline First, Always

A The teacher app must function completely without internet. Diary submission, attendance, student list — all must work offline. Sync is a background process, never a blocker. Any feature that requires connectivity to function is not shippable.

1**L RLS Is Not Optional**

A Row-Level Security must be implemented at the PostgreSQL layer for every table containing user or student data.
V API-layer filtering is a supplement, never the primary security control. Test: terminate the API, query the DB directly — data must still be role-scoped.

2**L Raw Data Is Immutable**

A diary_entries and student_sessions records cannot be updated or deleted after 11:59PM on the day they were created.
V Corrections go in a separate corrections_log table with reason and approver. The original record is never touched.

3**L The 2-Minute Diary**

A Every UI sprint touching the diary entry screen must be benchmarked on a Tecno Spark with a test teacher. If the full diary cannot be completed in under 2 minutes, the sprint fails acceptance and is rejected.

4**L Computed Views Are Never Edited**

A computed_waec_meters and brain_maps are written only by the BullMQ computation engine. No endpoint, no admin panel, no migration may write directly to these tables. If the algorithm is wrong, fix the algorithm and re-run the job.

5**L Search Returns in Under 2 Seconds**

A Any search query — including the most complex district filter — must return results in under 2 seconds for datasets up to 50 schools and 25,000 students. This is a performance SLA enforced with automated load tests in CI/CD.

6**L Low-End Android Is the Target Device**

A Every mobile feature must be tested on a device with ≤2GB RAM, Android 10+, and a 3G connection. Features that only work on flagship phones are not shipped.

7

L WhatsApp Templates Need Pre-Approval

A No WhatsApp message can be sent to parents without a Meta-approved template. Template changes require
V re-submission (2–5 days). Build the template library first, get approvals, then build the sending logic.

8

L NDPR Compliance Is Non-Negotiable

A Student data is children's personal data under Nigerian law. Consent must be recorded before data is collected. Data
V must be encrypted at rest and in transit. WhatsApp message content purged after 18 months. A data processing
g agreement must exist with every third-party service touching student data.

L Ministry Sees No Individual Names

A No endpoint, no export, no dashboard feature at Ministry or District level may return individual student names as
V personally identifiable data. Aggregated, anonymised, or coded IDs only. Exceptions require written PS approval and are
1 logged in the audit trail.

0

L Lagos Colours Are Exact

A Blue: #0033A0 | Green: #008751 | Bright Green: #22C55E | Yellow: #FFCC00 | Red: #E30613. No variations, no
V 'close enough'. Brand integrity is a product requirement, not a design preference.

1

1

L This Document Is the Source of Truth

A If a feature is not in this document it is not in scope. If it is in this document it must be built exactly as described.
V Engineers who disagree with a specification raise it to the PM in writing before building — not after. Silent deviations are
1 rejected at code review.

2

• Acceptance Criteria — One Per Portal

How to use this section: Each item below is a complete user story acceptance test. A portal is not releasable until its acceptance criterion passes on a real device in a real network environment. No exceptions.

- ☐ **TEACHER APP:** A teacher with no training submits a complete diary (attendance + topic + score + 5 traits) in under 2 minutes on a Tecno Spark with no connectivity. Diary syncs automatically when WiFi is restored.
- ☐ **STUDENT PORTAL:** A JSS2 student completes a 10-question CBT practice session, sees instant per-question feedback, and their WAEC Readiness Meter updates in the following morning's computation run.
- ☐ **SCHOOL ADMIN:** A headteacher opens the dashboard and within 10 seconds identifies which teacher has the lowest compliance rate and sends them a nudge — without leaving the dashboard screen.
- ☐ **DISTRICT PORTAL:** A district officer searches 'Low Focus students in SS1 Physics' and receives a filtered list of student IDs, schools, and readiness scores in under 2 seconds.

- ☐ MINISTRY CENTER: The Permanent Secretary opens the dashboard and sees statewide readiness, district comparison, and top 3 problem subjects in under 10 seconds — no clicks required.
- ☐ PARENT WHATSAPP: At 6:05PM on a Friday, a parent receives a personalised WhatsApp report for their child, replies 'MORE', and receives a detailed subject breakdown within 30 seconds.
- ☐ BRAIN MAP: After 5 weeks of diary data, a student's Brain Map shows meaningful differentiation across the 5 cognitive domains, and the AI-generated teaching recommendation is specific and actionable.
- ☐ AI TUTOR: A student sends 'explain quadratic equations' to the WhatsApp tutor number and receives a clear, Lagos-contextualised explanation followed by a practice question within 15 seconds.
- ☐ SECURITY: A teacher with a valid JWT cannot retrieve diary entries, student profiles, or any data belonging to a different school — even with direct API calls.
- ☐ PERFORMANCE: The full system (all portals active, 50 schools, 18,500 students) maintains < 2s API response time at p95 under simulated daily load tested in the CI/CD pipeline.

"No child is dull."

Build accordingly.

Eko Excel OS — Engineering Constitution v1.0 — Lagos State Ministry of Education — 2025